

# Electricidad I:

## *Impedancia y admitancia*

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4 de octubre de 2026



# Impedancia

Impedancia

$$\mathbf{Z} = \frac{\mathbf{v}}{\mathbf{i}}$$

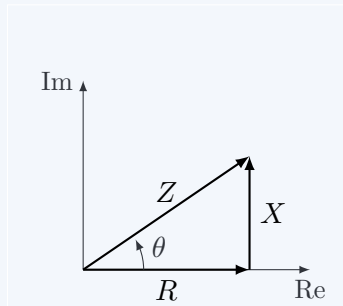
$$\mathbf{Z} = R + jX = |\mathbf{Z}| \angle \theta$$

$$|\mathbf{Z}| = \sqrt{R^2 + X^2} \quad \theta = \arctan\left(\frac{X}{R}\right)$$

$$R = |\mathbf{Z}| \cos \theta \quad X = |\mathbf{Z}| \sin \theta$$

donde  $R$  es la resistencia y  $X$  es la reactancia, ambas en ohms ( $\Omega$ )

Ajustar el ángulo según el cuadrante: si  $R < 0$ , sumar  $180^\circ$  cuando  $X \geq 0$  o restar  $180^\circ$  cuando  $X < 0$ .



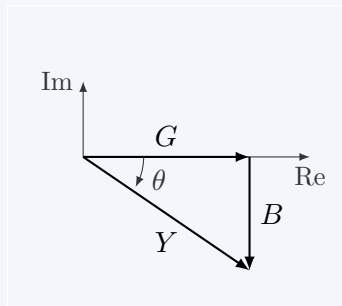
# Admitancia

Admitancia

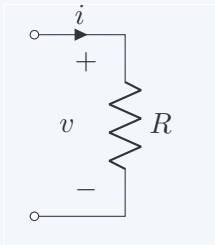
$$\mathbf{Y} = \frac{i}{v}$$

$$\mathbf{Y} = G + jB = \frac{1}{|\mathbf{Z}|} \angle -\theta$$

donde  $G$  es la conductancia y  $B$  es la susceptancia, ambas en siemens (S)



# Circuito resistivo



$$i = I \sin(\omega t + \phi)$$

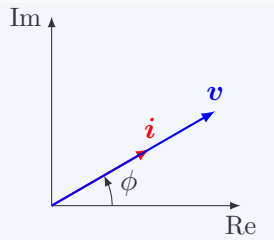
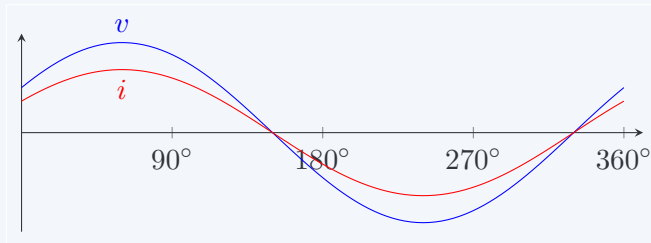
$$v = iR = RI \sin(\omega t + \phi)$$

$$i = I \angle \phi$$

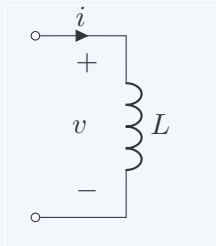
$$v = Ri = RI \angle \phi$$

$$Z = \frac{v}{i}$$

$$Z = R + j0 = R \angle 0^\circ$$



# Circuito indutivo



$$i = I \sin(\omega t + \phi)$$

$$v = L \frac{di}{dt} = \omega L I \sin(\omega t + \phi + 90^\circ)$$

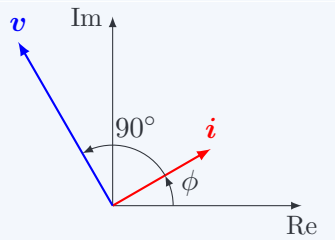
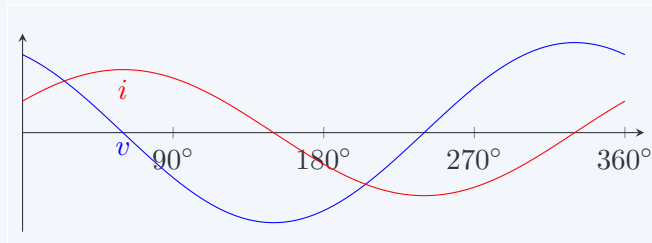
$$i = I \angle \phi$$

$$v = j\omega L i = \omega L I \angle \phi + 90^\circ$$

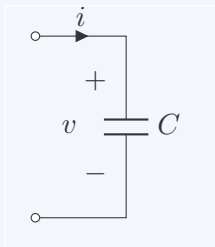
$$Z = \frac{v}{i}$$

$$Z = 0 + j\omega L = \omega L \angle 90^\circ$$

$$X_L = \omega L$$



# Circuito capacitivo



$$v = V \sin(\omega t + \phi)$$

$$i = C \frac{dv}{dt} = \omega CV \sin(\omega t + \phi + 90^\circ)$$

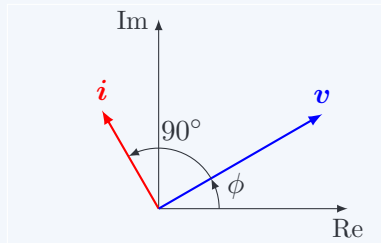
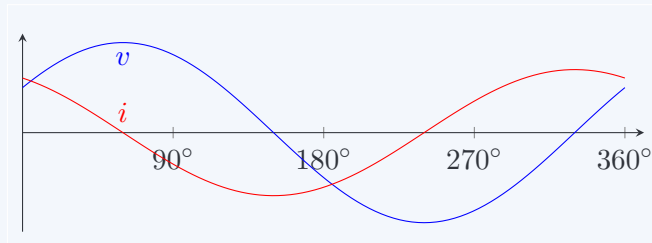
$$v = V \angle \phi$$

$$i = j\omega C v = \omega CV \angle \phi + 90^\circ$$

$$Z = \frac{v}{i}$$

$$Z = 0 + \frac{1}{j\omega C} = \frac{1}{\omega C} \angle -90^\circ$$

$$X_C = -\frac{1}{\omega C}$$



# Impedancias en serie

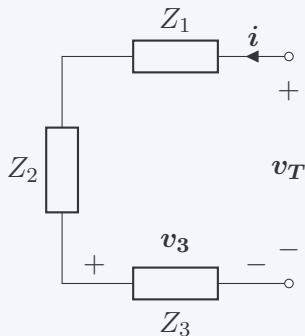
Impedancia equivalente

$$Z_{eq} = Z_1 + Z_2 + Z_3$$

Divisor de voltaje

$$v_n = v_T \cdot \frac{Z_n}{Z_1 + Z_2 + \dots + Z_n}$$

$$v_3 = v_T \cdot \frac{Z_3}{Z_1 + Z_2 + Z_3}$$



# Impedancias en paralelo

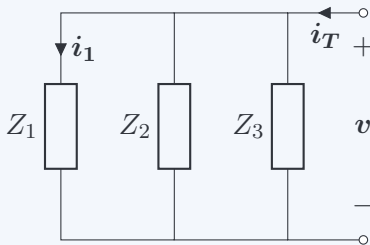
Impedancia equivalente

$$\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}$$

Divisor de corriente

$$i_n = i_T \cdot \frac{\frac{1}{Z_n}}{\frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_n}}$$

$$i_1 = i_T \cdot \frac{\frac{1}{Z_1}}{\frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}}$$



# Impedancias en paralelo: dos impedancias

Impedancia equivalente

$$Z_{eq} = \frac{Z_1 \cdot Z_2}{Z_1 + Z_2}$$

Divisor de corriente

$$i_1 = i_T \cdot \frac{Z_2}{Z_1 + Z_2}$$

